

ECE 3436 Exam #3

SPR, Encode, GPIO and SYSTICK

EXAM RULES:

- DO NOT open or start the exam until told to do so by the instructor.
 - This exam is open-book and open-notes. You are also allowed to use computers and/or calculators. However, you are not allowed to communicate with other students.
 - If you have any questions, ask the instructor. You will not be given credit for work that is based on a wrong assumption.
 - Please sit separately, don't be too close to each other.
 - If you finish the exam before the time is up, raise your hand and the instructor will collect your exam. You may then quietly leave after your exam is submitted.
 - When time is called, you must stop all writing and put your pens/pencils down. Stay seated until all exams have been collected. The instructor will notify you when you may leave.
 - You **MUST** sign the academic honesty agreement below before taking the exam. Failure to do so will result in a **zero** on the exam.
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ACADEMIC HONESTY AGREEMENT:

I Understand and agree to abide by the provisions in the **University of Houston Undergraduate Academic Honesty Policy**. I promise that the work on this exam is my own and that I have received no help during this exam other than any assistance provided by the instructor and/or Tas in this course. I understand that academic honesty is taken very seriously, and in the cases of violations, penalties may include suspension or expulsion from the University of Houston.

Name (print): _____

Signature: _____

Special Purpose Register

Use the following code and image (the codes are the same) on the following page to answer question 1-8. Be careful, the subroutine below is recursive (it calls itself).

Before the main code, we have defined the constant “Input” using “**Input DCD 4**”,
and have allocate space for the variable “Output” using “**Output SPACE 4**”

After compiling and initialization, the initial value in the Stack Pointer register (before running the main code) is 0x20000408, the address of the Start label is 0x00000284, and the value in LR register is 0xFFFFFFFF.

Hint: Be careful about the values in register R0 and R2.

```
AREA    |.text|, CODE, READONLY, ALIGN=2        ; Flash ROM

EXPORT  Start

Start      ; start executing here

    LDR    R1, =Input
    LDR    R0, [R1]
    MOV    R2, #1
    BL     Recursion
    LDR    R1, =Output
    STR    R2, [R1]
    NOP
    B      EndFunc

Recursion
    PUSH   {R0, LR}
    CMP    R0, #1
    BEQ    EndRecursion
    SUB    R0, #1
    BL     Recursion
EndRecursion
    POP    {R0, LR}
    MUL    R2, R0, R2
    BX     LR

EndFunc
    NOP
    NOP

ALIGN     ; make sure the end of this section is aligned
END       ; end of file
```

Registers

Register	Value
Core	
R0	0x00000003
R1	0x000002BC
R2	0x00000001
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20000400
R14 (LR)	0x000002A7
R15 (PC)	0x00000298
xPSR	0x21000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	20
Sec	0.00000125
FPU	

Disassembly

```

0x00000294 BF00 NOP
42: B EndFunc
43:
44: Recursion
0x00000296 E00B B 0x000002B0
45: PUSH {R0, LR}
0x00000298 B501 PUSH {r0,lr}
46: CMP R0, #1
0x0000029A 2801 CMP r0,#0x01
47: BEQ EndRecursion
0x0000029C D003 BEQ 0x000002A6
48: SUB R0, #1
0x0000029E F1A00001 SUB r0,r0,#0x01
49: BL Recursion
50: EndRecursion
0x000002A2 F7FFFFFF9 BL.W 0x00000298
51: POP {R0, LR}
0x000002A6 E8BD4001 POP {r0,lr}
52: MUL R2, R0, R2
0x000002AA FB00F202 MUL r2,r0,r2
53: BX LR

```

main.s Startup.s

```

33 Start ; start executing here
34
35 LDR R1, =Input
36 LDR R0, [R1]
37 MOV R2, #1
38 BL Recursion
39 LDR R1, =Output
40 STR R2, [R1]
41 NOP
42 B EndFunc
43
44 Recursion
45 PUSH {R0, LR}
46 CMP R0, #1
47 BEQ EndRecursion
48 SUB R0, #1
49 BL Recursion
50 EndRecursion
51 POP {R0, LR}
52 MUL R2, R0, R2
53 BX LR
54
55
56 EndFunc

```

Project Registers

1. What is the hexadecimal address of the next instruction to be executed?

Address: 0x_____

2. Is the stack currently empty? (Circle the correct answer.)

Yes No

3. What is the machine code (hexadecimal) for the instruction Line 45 `PUSH {R0, LR}`?

ARM machine code: 0x_____

4. At what memory address can you find the instruction Line 51 `POP {R0, LR}`?

Address: 0x_____

5. What will be the value stored (hexadecimal) in the PC immediately after the instruction on line 45 is executed? (Immediately after the moment shown in the image.)

Answer: 0x_____

6. What will be the value stored (hexadecimal) in the SP immediately after the instruction on line 45 is executed? (Immediately after the moment shown in the image.)

Answer: 0x_____

7. What will be the value stored (hexadecimal) in the LR immediately after the instruction on line 49 `BL Recursion` is executed?

Answer: 0x_____

8. What are the values in register R0 and R2 after running the code.

R0: 0x_____

R2: 0x_____

9. Please show what the following SRAM memory looks like after running the code. Assume the SRAM is cleared before running the code. Remember, the `POP` function will not clear the memory, it only changes the stack pointer, and the `POP {R0, LR}` instruction will push `LR` into the stack first.

Address	Values in the memory (A cell means a byte)							
0x200003E0								
0x200003E8								

0x200003F0								
0x200003F8								
0x20000400								
0x20000408								

General Purpose Input/Output (GPIO) Configuration

10. Which port clocks are activated by executing the following code? (Circle all that apply.)

PORTS: A B C D E F

```
LDR R1, =SYSCTL_RCGCGPIO_R
LDR R0, [R1]
ORR R0, R0, #0x3A
STR R0, [R1]
NOP
```

11. Which pins on **Port B** are configured to be analog pins after executing the code below? (Circle all that apply.)

PINS: 0 1 2 3 4 5 6 7

```
LDR R1, =GPIO_PORTB_AMSEL_R
LDR R0, [R1]
ORR R0, R0, #0x25
BIC R0, R0, #0xDA
STR R0, [R1]
NOP
```

12. What is the hexadecimal address for **Port B**'s Alternative Function Select (GPIOAFSEL) register? (Assume APB.)

Answer: 0x_____

13. Write the code to configure bit 0-3 for **Port C** as inputs and bits 4-7 as outputs. You are not allowed to change bits 8-31 of the configuration register. (Do not use labels in your code. Use the actual address for the configuration register.)

14. Write the code to enable the **pull-up** register on pins 0-3 on **Port D** (APB bus). Do not change the values for any other pins besides 0-3. (Do not use labels in your code. Use the actual address for the configuration register.)

15. If you just want to access the bits 2-5 (start from 0) of the **Port A** GPIO Data (GPIODATA) register, what bit-specific address you should use. (Assume APB.)

Address: 0x_____

16. Assume that initially all LEDs on **Port F** of the TI LaunchPad are off. After the code below is executed, what is the state of the three built-in LEDs connected to **Port F**? (Circle the correct answers.)

Red LED is:	on	off	don't know
Blue LED is:	on	off	don't know
Green LED is:	on	off	don't know

```
LDR  R1, =0x40025028
MOV  R0, #2_00110
STR  R0, [R1]
```

Clock System

17. The TM4C123 microcontroller system clock can be configured to run at a speed of 20 MHz. (Circle the correct answer.)

True False

18. When no clock configuration is performed, the TM4C123 microcontroller system clock runs at 16 MHz.

True False

19. Which of the following are **configuration registers** used to set up the system clock?
(Circle all correct answers)

RCC RCC2 RCC3 XTAL MOSC

SysTick Timer

20. What is the value will be in the NVIC_ST_RELOAD_R register after the following instruction.

```
MOV R0, #23DD0000
SysTick_Wait
LDR R1, =NVIC_ST_RELOAD_R
STR R0, [R1]
```

Answer: 0x_____

21. If the value in the SysTick Timer reload register is 0x1600, please complete the following sequence to show the values in the SysTick Timer current value register along system clock cycles.

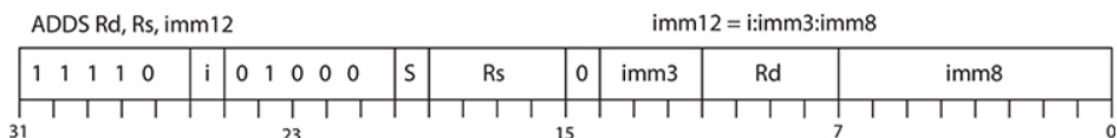
Clock	...	n	$n+1$	$n+2$	$n+3$	$n+4$	...
Values	...	2	1				...

22. Assume the following values are read from NVIC_ST_CTRL_R register, please circle the value shows the counter has ever hit zero.

0x00010005 0x00000005

Encoding instructions

23. The ADD(S) instruction will be encoded into a 32-bit machine code as shown in the following.



Please derive the machine code of instruction : `ADDS R1, R2, #0x33`

Instruction Bits (Most significant bytes)															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Instruction Bits (Least significant bytes)															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Answer: 0x_____

Program Design

24. Assume you want to design a digital lock using TM4C123 microcontroller, denote the input of SW2 is “0” and the input of SW1 is “1” (A rising edge or a falling edge means one press or input). After the system is reset, the red LED is on. If the input is correct (let’s assume the input is 10101110), turn on green LED and turn off red LED. Otherwise, the red LED will still be on.

Please use descriptive language to write down your design, the function of each part, and approach of implementation. You don’t need to write the code.